Tutorial 3

The scope of this tutorial Session is to help students understanding different data types related to Usability Testing.

Task 1 Identifying Data Types

Work in pairs considering the following usability testing scenarios and identifying which data type each example represents (nominal, ordinal, interval and ratio).

Scenario1: Testing an E-commerce Checkout Process

Description: Test the checkout process of an online store to assess ease of use, speed, and user satisfaction.

Data Collected:

- **Nominal Data**: Device type used for testing (e.g., "desktop," "tablet," "mobile").
- **Ordinal Data**: User satisfaction rating for the checkout process (e.g., "Very dissatisfied," "Dissatisfied," "Neutral," "Satisfied," "Very satisfied").
- **Interval Data**: Confidence rating on completing the checkout task (e.g., scale from 1 to 5, where 1 = "Not confident at all" and 5 = "Very confident").
- **Ratio Data**: Time taken to complete the checkout process (in seconds), number of clicks needed to complete the checkout.

Scenario 2: Evaluating a Mobile App's Navigation Menu

Description: Assess the ease of navigating through different sections of a mobile app via its main navigation menu.

Data Collected:

- **Nominal Data**: Type of error encountered (e.g., "misclick," "wrong section opened," "unintended swipe").
- **Ordinal Data**: Difficulty rating for finding specific menu items (e.g., "Very easy," "Easy," "Moderate," "Difficult," "Very difficult").
- **Interval Data**: User satisfaction with the menu layout (e.g., 1–7 scale, where 1 = "Very dissatisfied" and 7 = "Very satisfied").

- **Ratio Data**: Number of menu taps required to reach a specific section, time taken to locate and access a section (in seconds).

Scenario 3: Testing the Search Function on a Knowledge Base Website

Description: Test how easily users can find information on a knowledge base or help center website using the search function.

Data Collected:

- **Nominal Data**: Search filters used by the user (e.g., "articles," "videos," "user guides").
- **Ordinal Data**: User rating of search relevance (e.g., "Not relevant," "Somewhat relevant," "Moderately relevant," "Very relevant").
- **Interval Data**: Perceived ease of use for the search function (e.g., 1–10 scale, where 1 = "Very difficult" and 10 = "Very easy").
- **Ratio Data**: Time taken to find desired information (in seconds), number of search attempts made by the user.

Scenario 4: Testing the Signup Process on a Web Application

Description: Evaluate the user experience during the account signup process on a web app.

Data Collected:

- **Nominal Data**: User demographic information (e.g., "age group," "location").
- **Ordinal Data**: User rating of signup form clarity (e.g., "Very unclear," "Unclear," "Neutral," "Clear," "Very clear").
- **Interval Data**: Satisfaction with the signup experience (e.g., scale from 1 to 5).
- **Ratio Data**: Total time taken to complete signup (in seconds), number of fields that had to be re-entered due to errors.

Scenario 5: Testing an Interactive Dashboard on a SaaS Platform

Description: Assess the usability of a dashboard feature that provides analytics and insights for users.

Data Collected:

- **Nominal Data**: Types of data filters applied by the user (e.g., "date range," "location," "category").
- **Ordinal Data**: User rating of dashboard data visualization clarity (e.g., "Very unclear," "Unclear," "Neutral," "Clear," "Very clear").
- **Interval Data**: Satisfaction with the ease of customizing the dashboard (e.g., 1–10 scale).

- **Ratio Data**: Number of customizations made to the dashboard, time spent customizing settings (in seconds).

Task 2

Complete a Usability Testing Task and identify the test data type,

Work in pairs.

Take part in a Usability testing study and help evaluate the usability of facilities offered via the University of Westminster Website. To take part in the study access the following URL: **https://t.maze.co/194939830** and answer the questions.

This study assumes that the participants are current students at the University of Westminster, and should be familiar with the University Website.

For IIT ask students to use the IIT website and the system you use to access the Library. Use maze to create a survey, it is free to create an account and it will take you 10 minutes to create the survey. Alternatively, you could create a Google form. If you cannot do any of the two suggestions, the questions are provided below. Ask your students to complete the test using those questions assessing the IIT website.

The questions included in the test are the ones listed below.

Discuss with your tutor the following for each of the following questions:

[1] what type of data you collect?
[2] how would you code it to analyse it?
[3] What type of statistical analysis you could conduct?
[4] What would be an appropriate graph to visualise the results?

Q1: How often do you use the University Website? Only one selection is allowed.

- ○ Several times per day
- ○ Once a day
- ○ Three times a week
- ○ Once a week
- ○ Never
- ○ Other

[1] what type of data you collect? – Ordinal, categorical
[2] how would you code it to analyse it? A code from 1-6 for each selected option
[3] What type of statistical analysis you could conduct? simple descriptive statistics such as counts and frequencies. Additionally, you can use **percentiles**, **median**, **mode** and the **interquartile range** to summarise your data.
[4] What would be an appropriate graph to visualise the results? Pie Chart, Bar Chart

Q2 For what purpose do you use the University Website? Select all that apply.

- ☐ To access my email
- ☐ To access Blackboard
- ☐ To access my Timetable
- ☐ To access the Library
- ☐ To check the University News
- ☐ Other

[1] what type of data you collect? – Nominal, categorical
[2] how would you code it to analyse it? A code from 1-6 for each selected option
[3] What type of statistical analysis you could conduct? simple descriptive statistics such as counts and frequencies.
[4] What would be an appropriate graph to visualise the results? Pie Chart, Bar Chart

Q3 Access the Library and find an article on "Usability testing for Virtual Reality applications". Did you find a relevant article?

- ○ Yes
- ○ No

[1] what type of data you collect? – Nominal, categorical
[2] how would you code it to analyse it? A code from 1-2 for each selected option
[3] What type of statistical analysis you could conduct? simple descriptive statistics such as counts and frequencies.
[4] What would be an appropriate graph to visualise the results? Pie Chart, Bar Chart

Q4 How long it took you to find a relevant article. Provide your answer in seconds.

Type your answer

[1] what type of data you collect? – Ratio
[2] how would you code it to analyse it? Note the exact value
[3] What type of statistical analysis you could conduct? When you are dealing with continuous data, you can use the most methods to describe your data. You can summarise your data using percentiles, median, interquartile range, mean, mode, standard deviation, and range.
[4] What would be an appropriate graph to visualise the results? To visualise continuous (Interval/Ratio) data, you can use a histogram or a box-plot. With a

histogram, you can check the central tendency, variability, modality, and kurtosis of a distribution. Note that a histogram can't show you if you have any outliers. This is why we also use box-plots.

Q5 How clear was to find a relevant article?



(this is what the user sees)



(this is the data that the system collects)

[1] what type of data you collect? – Interval data
[2] how would you code it to analyse it? A code from 1-5 for each selected option representing the intervals of the scale
[3] What type of statistical analysis you could conduct? When you are dealing with continuous data, you can use the most methods to describe your data. You can summarise your data using percentiles, median, interquartile range, mean, mode, standard deviation, and range.
[4] What would be an appropriate graph to visualise the results? To visualise continuous (Interval/Ratio) data, you can use a histogram or a box-plot.
With a histogram, you can check the central tendency, variability, modality, and kurtosis of a distribution.

Q6 Provide the citation of the article that you found on "Usability testing for Virtual Reality applications", using the correct citation.

[1] what type of data you collect? – Nominal, categorical
[2] how would you code it to analyse it? A code from 1-2 for correct completion, no completion

[3] What type of statistical analysis you could conduct? simple descriptive statistics such as counts and frequencies.

[4] What would be an appropriate graph to visualise the results? Pie Chart, Bar Chart

Q7 How long it took you to find the correct citation? Provide your answer in seconds.

[1] what type of data you collect? – Ratio

[2] how would you code it to analyse it? Note the exact value

[3] What type of statistical analysis you could conduct? When you are dealing with continuous data, you can use the most methods to describe your data. You can summarise your data using percentiles, median, interquartile range, mean, mode, standard deviation, and range.

[4] What would be an appropriate graph to visualise the results? To visualise continuous (Interval/Ratio) data, you can use a histogram or a box-plot. With a histogram, you can check the central tendency, variability, modality, and kurtosis of a distribution. Note that a histogram can't show you if you have any outliers. This is why we also use box-plots.

Discuss with your tutor the following questions:

- Is the test trying to identify:
  - Behavioural or Attitudinal data?
  - Qualitative or Quantitative data?

Answer

- We collect behavioural data as all the questions assume that the participants use the website and the data we collect is about user behaviour and performance
- The only Attitudinal data that we collected is Q5 How clear was to find a relevant article? Because here we collect data about how users feel.

- The ratio data (Q4 and Q7) is clearly quantitative. The other questions that we collect are qualitative, but they can be easily coded and converted to quantitative to be analysed. The only purely qualitative data we collect is when we ask the users to provide their answers. This data could be turned into quantitative if we check for repetitions in the provided answers.

- Could you run the test:
  - Moderated or Unmoderated?

- Remote or in-person?

Answer

- Moderated or Unmoderated?

  We run the study as unmoderated as we are not next to the participants assisting them during the process.

- Remote or in-person?

  We run the study as remote as we give users a URL directing them to complete tasks and recording their behaviour

Task 3 Collecting Quantitative (Nominal, Ordinal, Interval, Ratio Data) and Qualitative data

Continue working in pairs to simulate a study where you record quantitative **nominal**, **ordinal, interval** and **ratio** data, as well as qualitative data. For example device types (e.g., "desktop," "tablet," "mobile") and user feedback categories (e.g., "design," "speed," "usability").

Example Nominal Data Collection:

- **Example**: Ask participants to simulate a study where they record device types (e.g., "desktop," "tablet," "mobile") and user feedback categories (e.g., "design," "speed," "usability").
- **Activity**: Participants will design a basic data entry form or digital tool that allows for selecting nominal categories during testing.

Example Ordinal Data Collection:

- **Example**: Explain ordinal scales in usability testing (e.g., rating task difficulty as "easy," "moderate," "difficult").
- **Activity**: Participants will create an ordinal scale for task difficulty and a Likert scale for user satisfaction (e.g., 1–5 or 1–7 scale), then practice recording mock data.

Example Interval Data Collection:

- **Example**: Show how to collect interval data through System Usability Scale (SUS) scores or user confidence ratings on a 1–5 scale.
- **Activity**: Participants will practice filling out mock SUS scores for a sample interface and calculate averages, discussing how this can reflect user perceptions across tasks or versions.

Example Ratio Data Collection:

- **Example**: Demonstrate collecting time-based data for task completion or number of clicks to complete a task.
- **Activity**: Participants will use a timer or logging tool to record task completion time, and count mock clicks to complete a task. Discuss the importance of accuracy with ratio data due to its quantitative nature.

Here is an example created in smartsurvey. Complete it and they use smartsurvey to create your own survey

https://www.smartsurvey.co.uk/s/MO86TW/

Discuss the type of data which is collected in this example.

Create your own survey for a scenario of your choice and post the link of your survey in the Padlet to get the rest of the class to provide data for your usability testing task.

Explore alternative online free tools you can use to create and share surveys. Propose those in the Padlet.

Task 4 Designing a Usability Test with Mixed Data Types

Work in groups of 4 (try to work with different people to the ones you worked last week) Split participants into small groups and give each group a sample product or interface to test.

Design a usability test that collects nominal, ordinal, interval, and ratio data for each of the following scenarios:

Each group should specify:

The data type they'll collect (e.g., error type, task satisfaction, SUS scores, task completion time).

The tool or method they'll use for each data type.

Share your study plan to the Padlet and each group presents their test plan, explaining why each data type is relevant and how it supports their usability objectives.

Scenario 1: Testing a Social Media App's Profile Setup

- **Description**: Users are asked to create a profile on a new social media app and customize their settings.
- **Data Collected**:
  - **Nominal Data**: Profile picture choices (e.g., "uploaded photo," "default avatar").
  - **Ordinal Data**: Satisfaction with customization options (e.g., "Very dissatisfied" to "Very satisfied").
  - **Interval Data**: Confidence level in completing profile setup successfully (1–5 scale).
  - **Ratio Data**: Time taken to complete the profile setup (in seconds).
  - **Qualitative Data**: User feedback on which settings were difficult or intuitive to configure and reasons for satisfaction/dissatisfaction.

Scenario 2: Testing a Website's Search Function

- **Description**: Users are asked to find specific information on a website using the search function, such as locating a product or article.
- **Data Collected**:
  - **Nominal Data**: Search filter used (e.g., "articles," "products," "guides").
  - **Ordinal Data**: Relevance rating of search results (e.g., "Not relevant," "Somewhat relevant," "Moderately relevant," "Very relevant").
  - **Interval Data**: User rating of ease of use for the search bar (e.g., 1–10 scale).
  - **Ratio Data**: Number of search attempts or queries made to find the target information.
  - **Qualitative Data**: User comments on search experience, such as what they found confusing or particularly helpful in the search process.

Scenario 3: Testing the Checkout Process on an E-commerce Site

- **Description**: Users are tasked with purchasing an item from an e-commerce site, moving through the cart, checkout, and payment processes.
- **Data Collected**:
  - **Nominal Data**: Payment method selected (e.g., "credit card," "PayPal," "bank transfer").
  - **Ordinal Data**: Difficulty rating for each step in the checkout process (e.g., "Very easy" to "Very difficult").
  - **Interval Data**: Satisfaction rating with the overall checkout process (e.g., 1–5 scale).

- **Ratio Data**: Total time taken to complete the checkout process (in seconds), number of clicks required to finish the transaction.
- **Qualitative Data**: Open-ended feedback on pain points, particularly challenging steps, or general thoughts on the checkout flow.

Scenario 4: Testing Navigation in a Mobile Banking App

- **Description**: Users are instructed to locate their recent transactions and check their account balance in a mobile banking app.
- **Data Collected**:
  - **Nominal Data**: Type of navigation option used (e.g., "menu bar," "search," "shortcut button").
  - **Ordinal Data**: User rating of task difficulty (e.g., "Very easy," "Easy," "Moderate," "Difficult," "Very difficult").
  - **Interval Data**: Confidence in navigating through the app (1–10 scale).
  - **Ratio Data**: Number of taps/clicks to reach the desired information, time taken to complete navigation tasks (in seconds).
  - **Qualitative Data**: User feedback on navigation intuitiveness, any challenges in finding information, and suggestions for improving navigation flow.

Scenario 5: Testing a Task Management Software's Task Creation Feature

- **Description**: Users are asked to create a new task, set deadlines, and assign team members in a task management software tool.
- **Data Collected**:
  - **Nominal Data**: Type of task created (e.g., "personal," "team," "project-based").
  - **Ordinal Data**: Rating of ease of setting deadlines and assigning team members (e.g., "Very difficult" to "Very easy").
  - **Interval Data**: Satisfaction with task creation options (1–7 scale).
  - **Ratio Data**: Number of fields completed or modified in the task setup, total time to create and configure a task (in seconds).
  - **Qualitative Data**: Comments on usability, such as how users felt about task configuration options, whether instructions were clear, and suggested feature improvements.